

Machine Learning Practice Worksheet

House Price Prediction & Classification Practice

Dataset: `house_price_prediction.csv`

Data Exploration & Preprocessing

1. Load the dataset.

Q1. Explore the Dataset

Write the code to:

- Display the first five rows.
 - Find the number of rows and columns.
 - Check the data types.
 - Display summary statistics.
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Q2. Data Cleaning

Answer the following:

1. Does the dataset contain any missing values?
 2. Which column will be your target variable?
 3. Which columns are features?
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Q3. Feature Engineering

Create a new feature named:

Price_per_sqft

Formula:

Price_USD / Area_sqft

Display the first five values.

Q4. Feature Selection

Select the following features:

- Area_sqft
- Bedrooms
- Bathrooms
- Age_years
- Garage

Store them in **X** and store **Price_USD** in **y**.

Linear Regression

Q5. Split the Dataset

Split the data into:

- 80% Training
 - 20% Testing
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Q6. Train a Linear Regression Model

Use Linear Regression to train the model.

Predict prices for the testing data.

Q7. Model Evaluation

Calculate:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- R^2 Score

Write one sentence describing your model's performance.

Q8. Make a Prediction

Predict the price of a house with:

- Area = 2200 sqft
- Bedrooms = 4
- Bathrooms = 3
- Age = 6 years
- Garage = Yes

Write the predicted price.

Polynomial Regression

Polynomial Regression can capture curved relationships.

Q9. Train a Polynomial Regression model with **degree = 2**.

Answer:

1. Did the R^2 score improve compared to Linear Regression?
 2. Which model performed better?
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Q10. Logistic Regression

Train a Logistic Regression model using the same features.

Calculate the model accuracy.

Write the accuracy score.

Great job! You have now practiced the complete beginner Machine Learning workflow—from understanding ML concepts to building both Regression and Classification models.